

[INCHEON KORMARINE CONFERENCE 2026]



AI and Robotic Technologies for Production Innovation in Shipyards' Subcontractors

2026. 06.

Digital Manufacturing Innovation Division Song ChangSub



중소조선연구원
Research Institute of Medium & Small Shipbuilding



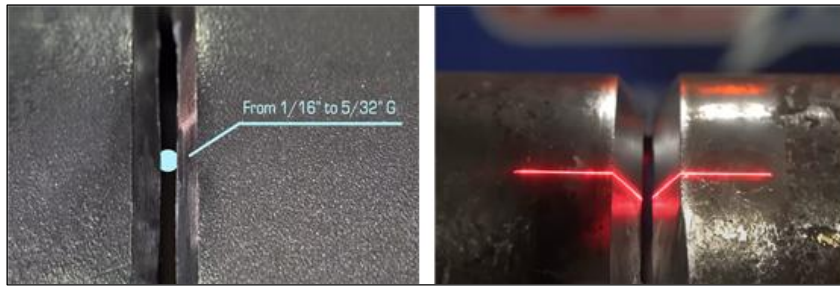
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- II AI Tech for Welding
- III Cases of Application
(The Project of Smart Production Innovation
for Small and Medium Shipbuilding)



1. Challenges in Shipbuilding

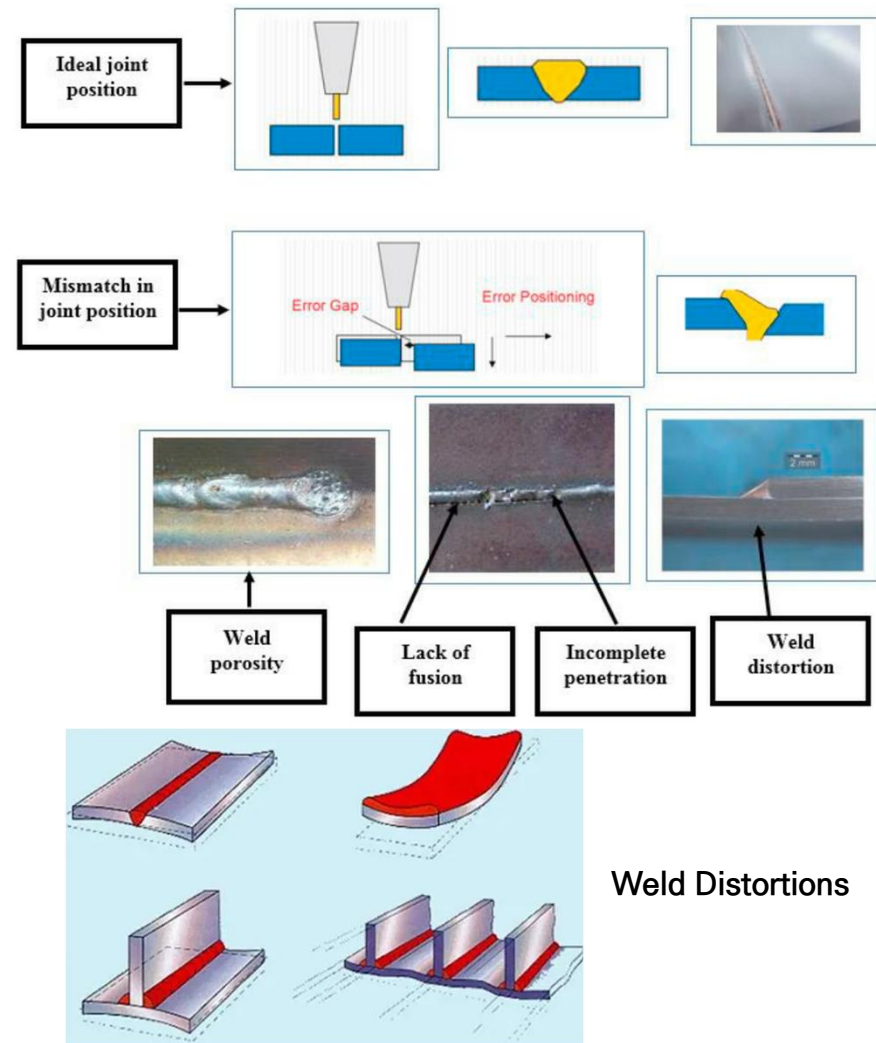
Challenges in Applying Robotics & Automation to Shipbuilding



Cutting Tolerances



Large & Irregular



Weld Distortions

Reference : Survey on AI applied in welding: A future scenario of the influence of AI on technological, economic, educational and social changes (Procedia Manufacturing, 2019)

Challenges in Applying Robotics & Automation to Shipbuilding



Narrow Workspaces / Complex & Diverse Structures



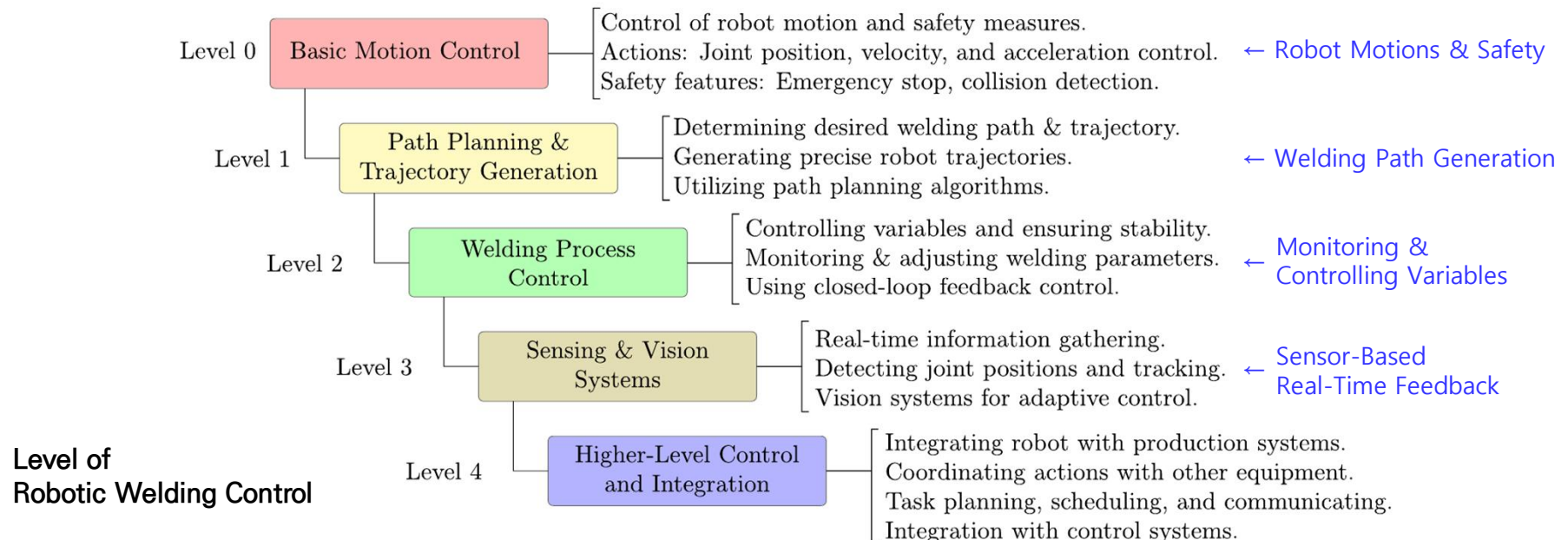
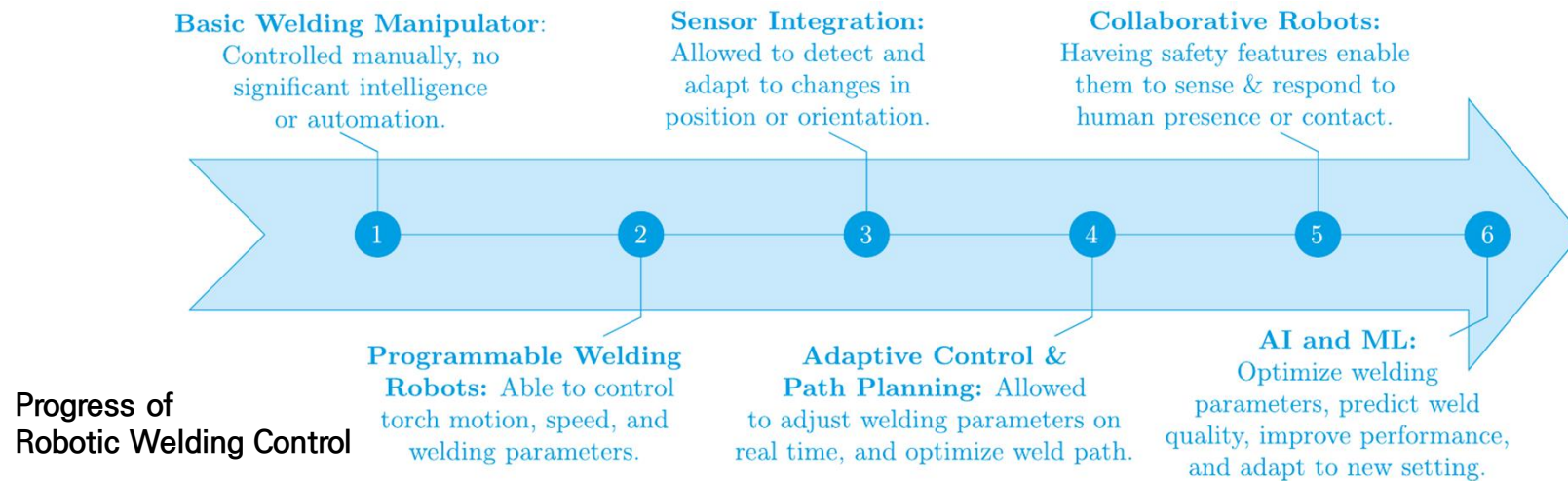
3D Shapes with Various Thickness & Materials



2. AI Tech for Welding

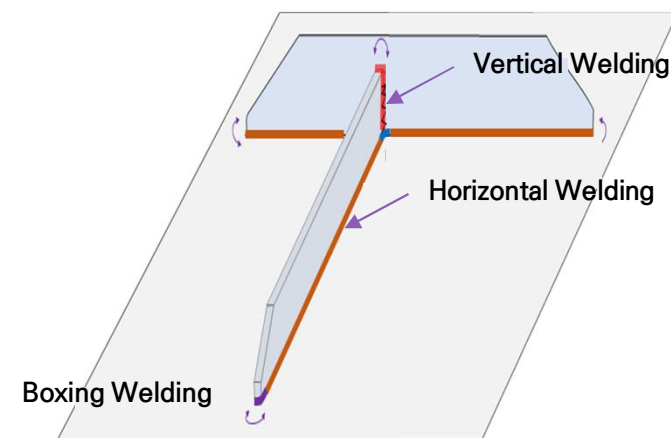
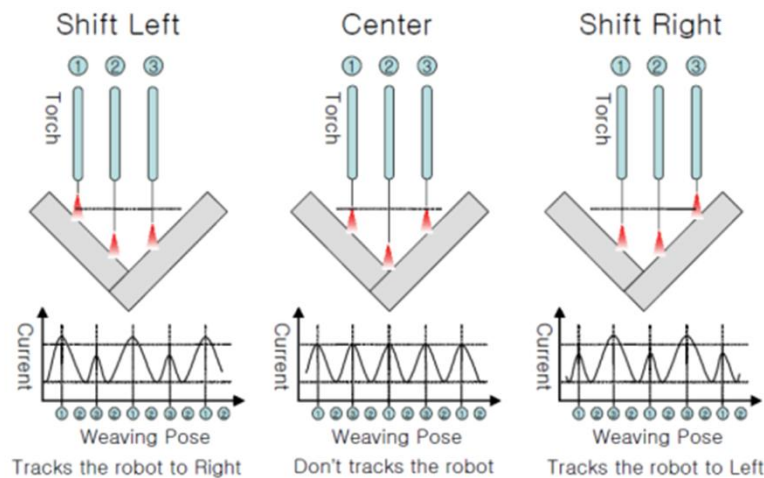
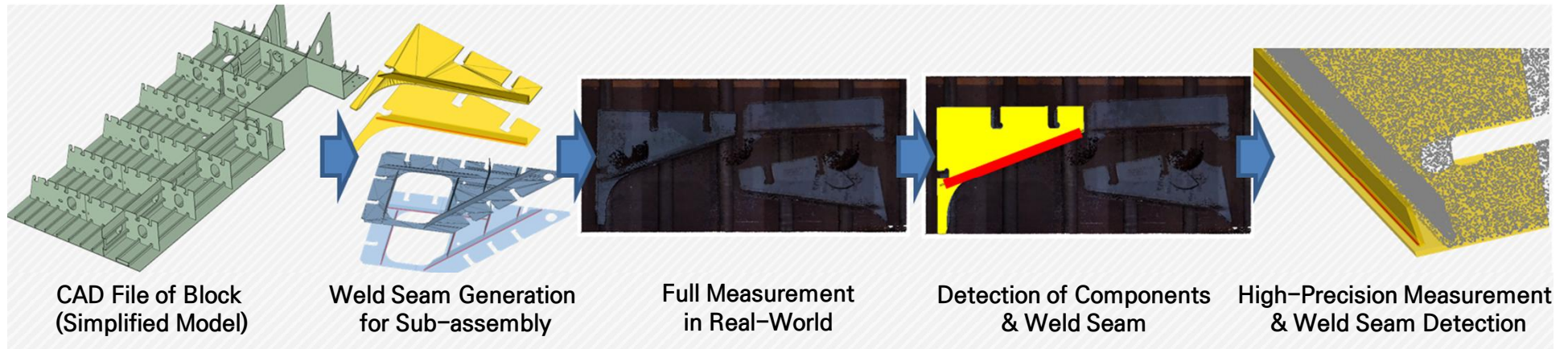
Progress of Welding Automation

Reference : Advancements in control systems and integration of artificial intelligence in welding robots: A review (Jalal Taheri Kahnamouei , 2024)



Automated Hull Assembly Technology

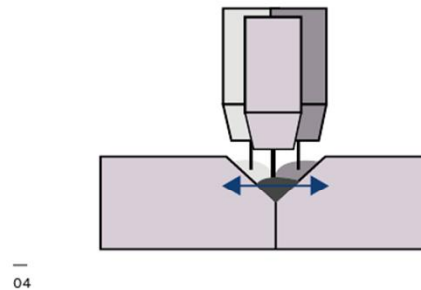
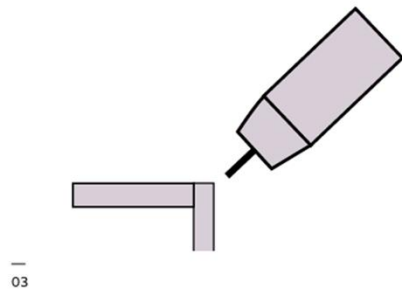
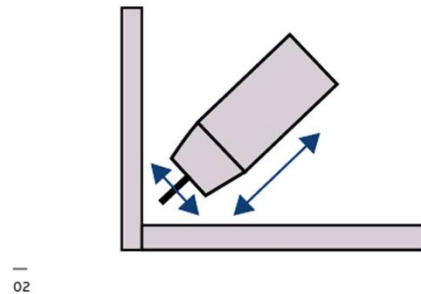
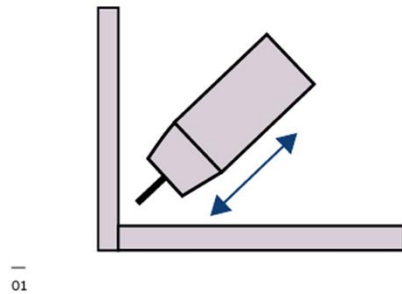
Welding



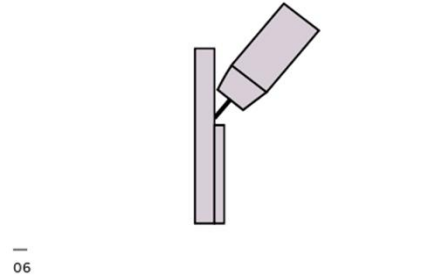
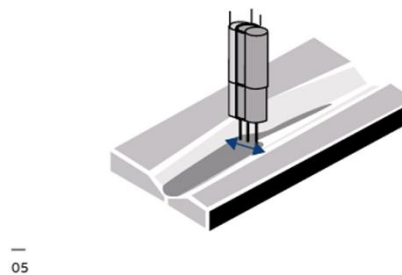
〈 Seam Tracking by Weld-Current Sensing / Welding Positions 〉

Automated Hull Assembly Technology

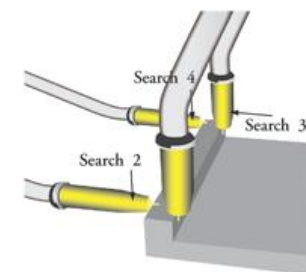
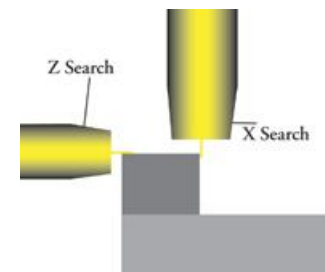
Welding



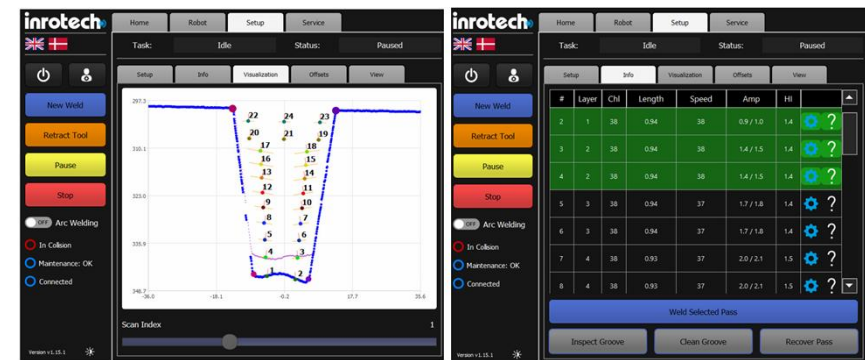
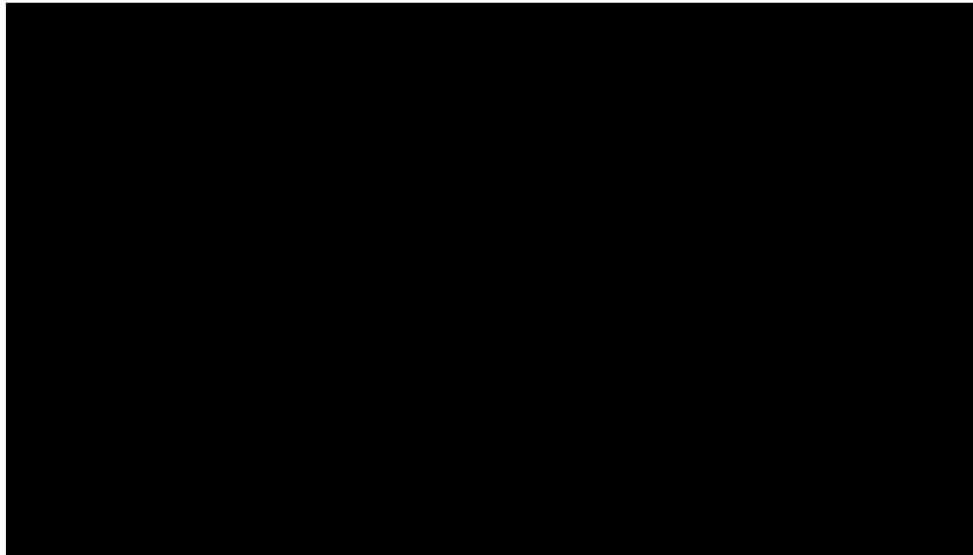
Tracking modes - Advanced



1. Height Measurement : Touch Sensor
2. Weld Centerline : Arc Sensor via Weaving
3. Corner Centerline : Arc Sensor via Weaving
4. Multi-Pass Weld Centerline : Arc Sensor via Weaving + Current Sensor
5. Non-uniform Groove Gap (Tolerance) : Arc Sensor via Weaving + Speed Sensor
6. Lap-joint : Touch Sensor



Automation of Multi-Pass Welding



- Applications
 - Eco-Fuel Tanks, Transport Tanks
 - Offshore Wind Substructures



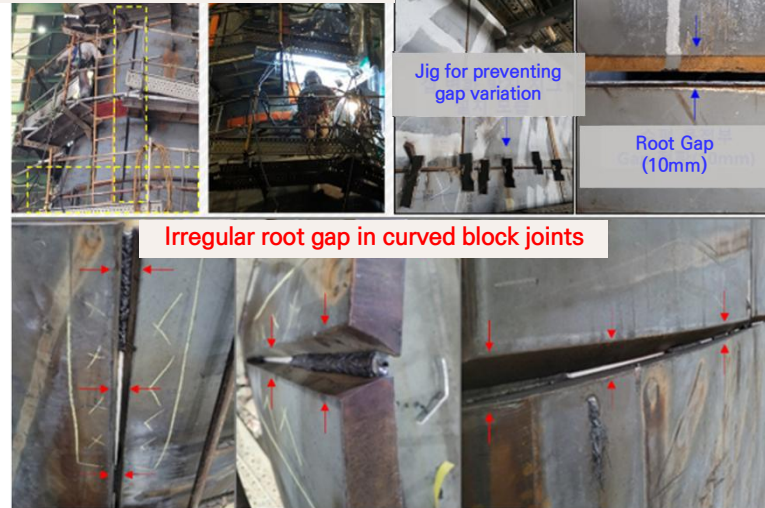
Automation of Curved Surface Welding

Representative Curved Block



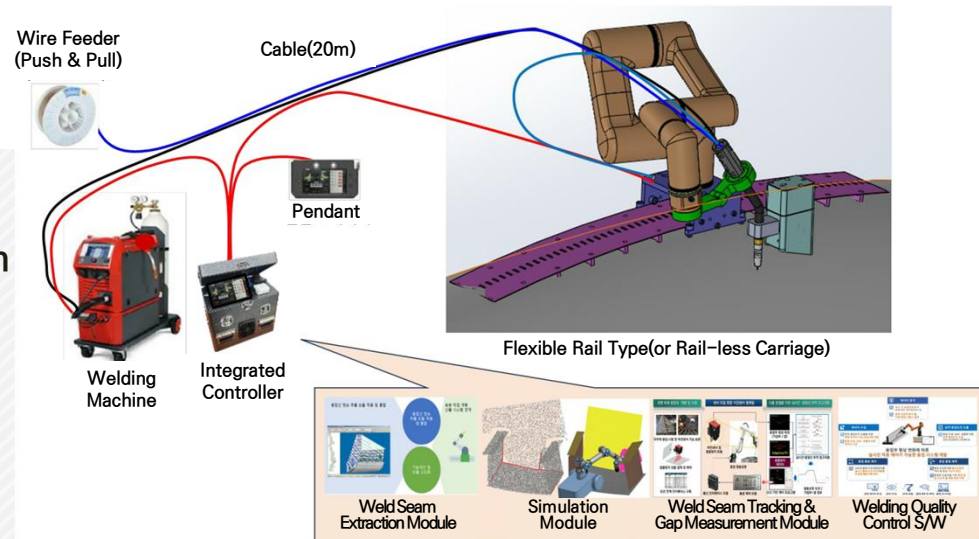
〈 Curved Block : Bow / Stern Sections of a Ship 〉

Curved block assembly by manual welding

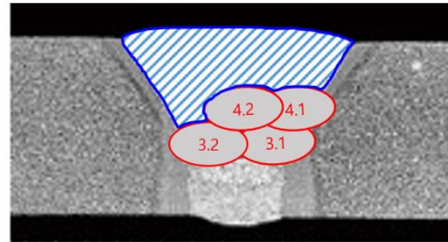


〈 Weld Joint Geometry Based on Curvature 〉

- Key Technologies
 - 3D Curved Surface Welding for Non-uniform Grooves (Horizontal & Vertical)
 - Auto-Detection of Non-uniform Gaps & Curvatures in Curved Blocks



Automation of Curved Surface Welding



Manual Manufacturing				Adaptive Manufacturing		
Input data	Algorithm	Output data		Input data	Algorithm	Output data
Visual Determination	Intuition & Experience	Required Weld Area for Pass	Welding Area Determination	Laser Scanning	Shape-aware Model	Required Weld Area for Pass
Required Weld Area	Welding Procedure Spec.	Combination of Welding Variables	Optimal Welding Variables	Required Weld Area	Regression Model	Combination of Welding Variables
Required Weld Area	Intuition & Experience	Welding Path	Optimal Position	Required Weld Area	Reinforcement Learning	Continuous Vector
Visual & Sound Data	Real-time Memory Recall	Welding Inspection	Quality Assessment	Time-series Welding Data	RNN & Kalman Filter	Anomaly Analysis Results

Pipe Welding Automation

〈 Development of Adaptive Manufacturing System of Pipe for Ship, 2024-2027 〉

Real-time Variables Control for Welding

❖ Input Data

- Welding variables data
- Expected rate: approx. 1K Row/sec per weld

❖ Output Data

- Welding variable (control) data

❖ Examples of Data

학습데이터								
특징	모재	와이어 제조사	와이어 직경	위빙폭	드웰타임	개선타입	개선종류	위빙 주파수
작업 고정	작업 고정	작업 고정	작업 고정	작업 고정	작업 고정	작업 고정	작업 고정	작업 고정
수기입력	○	○	○	○	○	○	○	○
비고	INT	INT	float	float	float	float	float	float

특징	갭	단차	용접속도	입력전류	측정전류*	전압	CTWD	시간	으로부터	양부
패스고정	가변	가변	가변	패스고정	가변	가변	가변	가변	가변	가변
○	센싱	센싱	자동	○	자동	자동	○	자동	자동	○
int	float	float	float	float	float	float	float	datetime	float	bool

*전류와 WFS는 상용 직준용계를
**속도를 시작으로 연산

INPUT 데이터								
특징	갭	단차	용접속도	입력전류	전압	모재	용가재 메이커	용가재 구경
작업 고정	가변	가변	가변	패스고정	가변	작업 고정	작업 고정	작업 고정
수기입력	센싱	센싱	자동	○	자동	○	○	○
비고	float	float	float	float	float	INT	INT	float

OUTPUT 데이터							
특징	입력전류	전압	CTWD	위빙폭	위빙 주파수	드웰타임	용접속도
패스고정	가변	가변	가변	작업 고정	작업 고정	작업 고정	가변
수기입력	○	○	○	○	○	○	자동
비고	float	float	float	float	float	float	float

Welding Pool/Defect Prediction

❖ Input Data

- 4 types of welding process data
- Video frame-based image data (1920x1080x3 RGB color channels)
- Each channel pixel value: integers 0 ~ 255

❖ Output Data

- Pass/Fail info & XY coordinate data

No.	전류(A)	전압(V)	송급량(mm/min)	유량(L/min)	이미지(640*480)
1	300	30	10	18	640*480
2	300	30	10	18	640*480
3	300	30	10	18	640*480
			10	18	640*480
			10	18	640*480
			10	18	640*480
			10	18	640*480
			10	18	640*480
			10	18	640*480
			10	18	640*480



〈 4 Types of Welding Process Data 〉

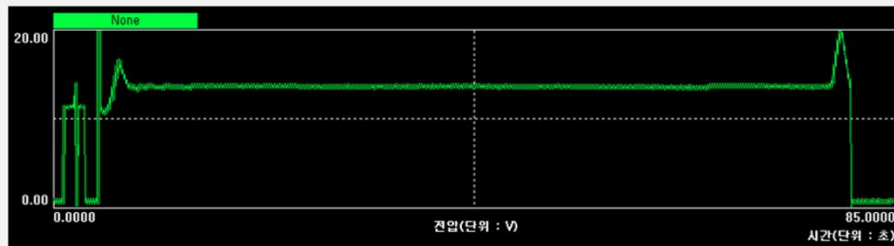
〈 Welding Image Data 〉

Pipe Welding Automation

Normal Bead & Data

❖ Bead Profile & Data under Normal Conditions

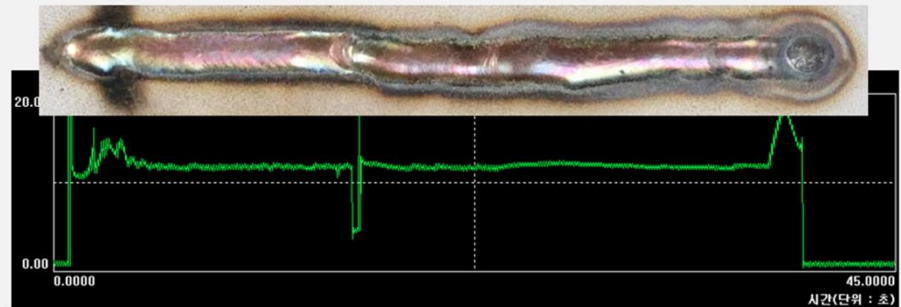
- A bead of consistent width is formed under normal conditions
- The collected signal maintains a steady range



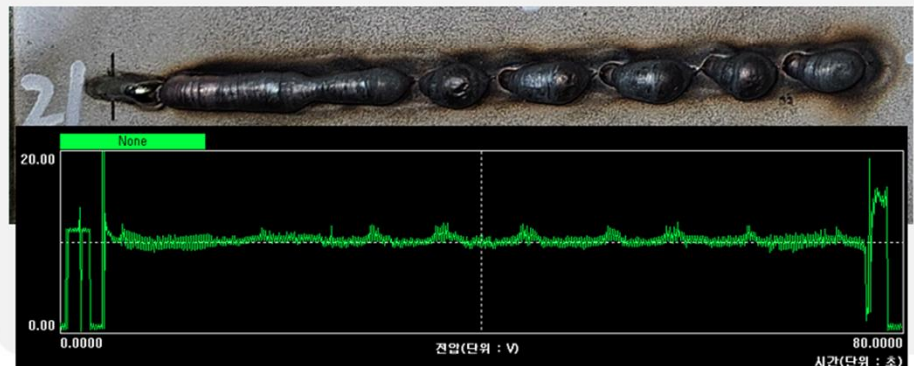
Defective Bead & Data

❖ Differences Based on Defect Data

- Sharp voltage fluctuations in the arc extinction area



- Voltage fluctuates on globular bead profile

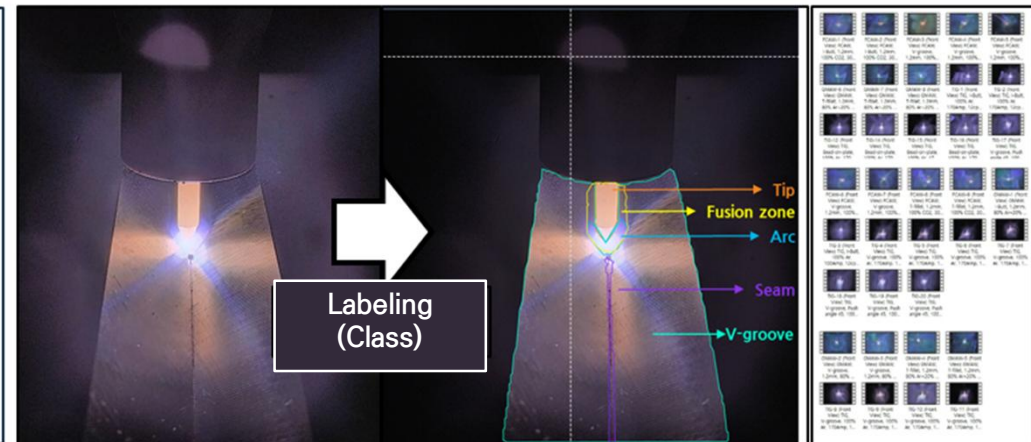


Pipe Welding Automation

Prototype of Welding Pool Extraction

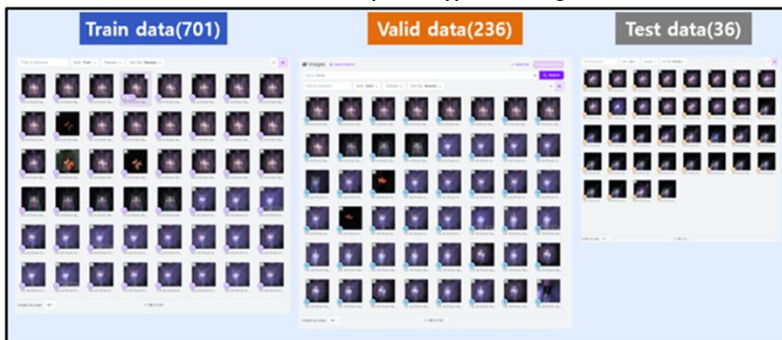
- ❖ Video data collection & generation by prototype welding process (GMAW, GTAW, FCAW)
- ❖ Splitting 1 raw video data (36s) into 72 image data
- ❖ Detection and prediction of weld pool arc and bead profile using YOLO algorithm

GTAW 영상촬영실험									
실험번호	Shading	Sense	전류	속도(mm/s)	Shading	Sense	전류	속도(mm/s)	후행
TIG-1	7								
TIG-2	8								
TIG-3	7								
GMAW 영상촬영실험									
실험번호	Shading	Sense	전류	속도(mm/s)	Shading	Sense	전류	속도(mm/s)	후행
GMAW-1	6								
GMAW-2	6								
GMAW-3	6								
GMAW-4	6								
GMAW-5	7								
GMAW-6	8								
GMAW-7	8								
GMAW-8	4								
GMAW-9									
GMAW-10									
GMAW-11									
GMAW-12									
FCAW 영상촬영실험									
실험번호	Shading	Sense	전류	속도(mm/s)	Shading	Sense	전류	속도(mm/s)	후행
I-Butt, 100% CO ₂									
FCAW-1	8	6	300	10	8	6	300	10	
FCAW-2	7	6	300	10	7	6	300	10	
V groove, 100% CO ₂									
FCAW-3	7	6	300	10	7	6	300	10	
FCAW-4	7	6	300	10	7	6	300	10	
FCAW-5	8	6	300	10	8	6	300	10	
FCAW-6	8	6	300	10	8	6	300	10	
FCAW-7	8	6	200	10	8	6	200	10	
T fillet, 100% CO ₂									
FCAW-8	7	6	300	10	7	6	300	10	
FCAW-9	8	6	300	10	8	6	300	10	
V groove - Drag angle : 45, 100% CO ₂									
FCAW-10									
FCAW-11									
FCAW-12									

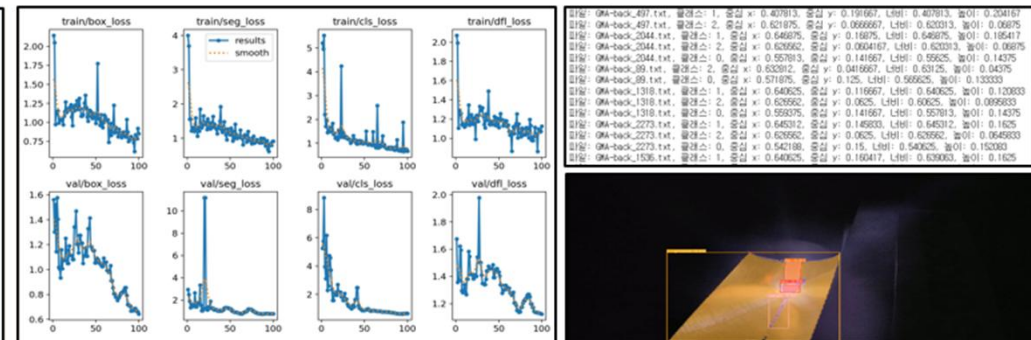


〈 Video Data Collection by Prototype Welding Process 〉

〈 Dataset Generation 〉



〈 Train/Valid/Test Data Splitting 〉



〈 YOLO Detection Algorithm and Acquisition of Video & Coordinate Data 〉



3. Cases of Application

(The Project of Smart Production Innovation for Small and Medium Shipbuilding)

- TAEJIN
- HJ Shipbuilding & Construction
- DAEYEON
- SUN-TECH

Automated Welding System for Ship Blocks (TAEJIN)

〈 Current Level: Limitations of Carriage Welding 〉

- Inability to operate multiple machines per person
- Requires arc monitoring and level of operator's skill
- Non-welded areas occur
- Limited welding positions
- Modifications required depending on the application



〈 Automation Solution: Collaborative Robot (Cobot) 〉

- **Multiple machine operation is available** / Portable
- No arc monitoring required (Database-driven welding)
- **Minimized non-welded areas** / No restrictions on welding positions
- **High welding quality (Arc sensing)**
- Enables welding with minimal training
(Low-skilled welders, including foreign and female workers)



〈 Overview of Ship Block Welding Site 〉



〈 Minimizing Non-welded Areas using Cobots 〉



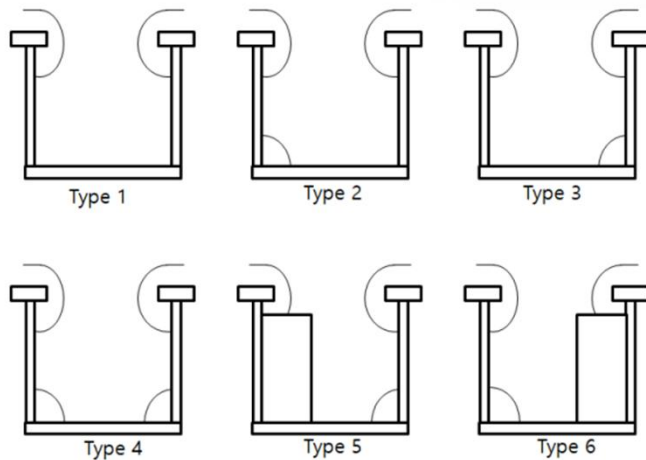
〈 Applicable Block Structures for the System 〉

Automated Welding System for Ship Blocks (TAEJIN)

1

Development of Automatic Workpiece Detection System

- o Camera mounted on the cobot recognizes type of workpieces
 - Boundary detection (using OpenCV edge detection functions)
- o Classifies 6 zones and Detects each space
 - Positions of Corner, Collar plates, and Presence of holes

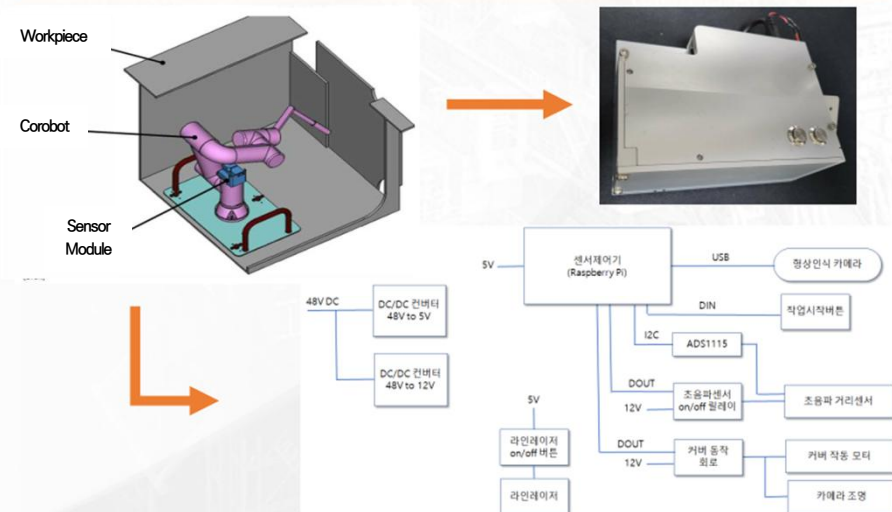


〈 Definition of Target Workpiece Shapes 〉

2

Development of Integrated Sensor Module

- o Workpiece size/distance measurement module
 - Measuring workpiece size by using ultrasonic sensor
- o Interfacing with the robot controller
 - Transmitting shape data to robot via TCP/IP Connection



〈 Integrated Sensor Module Layout & Connection Diagram 〉

Automated Welding System for Ship Blocks (TAEJIN)

3

Development of OLP(Off-Line Programming)

- o Program based on detected shape and measured dimensions
 - Develop standard programs for 6 different shape types
(Variables & coordinates are set according to the measured data)
- o Actual welding position: Precise measured by touch-sensing
 - Positions of Corner, Collar plates, and Presence of holes
(Auto-generated sensing/welding motion programs)

4

Programming for Co-robot Welding

- o (LUA language) Execute functions in the program according to the shape of the respective workspace
 - Receive shape data (location/dimension) by the sensor module
 - Position compensation based on generated coordinates
- o (Touch Sensing Sequence) Right vertical section → Right collar plate → Horizontal → Left collar plate → Left vertical section
- o (Welding Sequence) Horizontal → Right vertical weaving → Collar plate vertical weaving → Left vertical weaving



〈 Horizontal Welding Execution & Results 〉



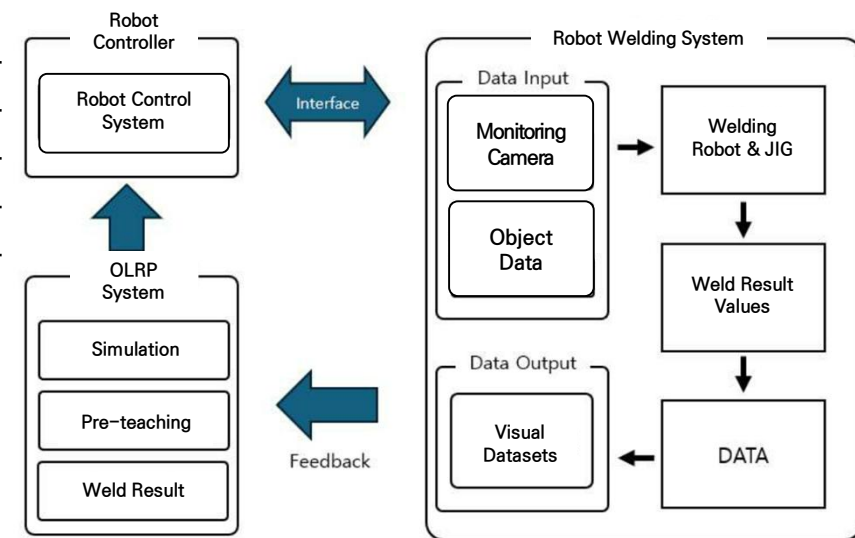
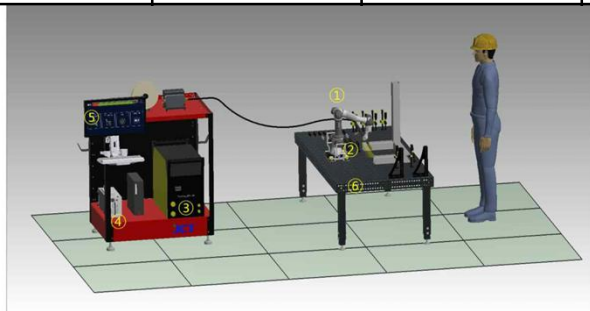
〈 Vertical Welding Execution & Results 〉



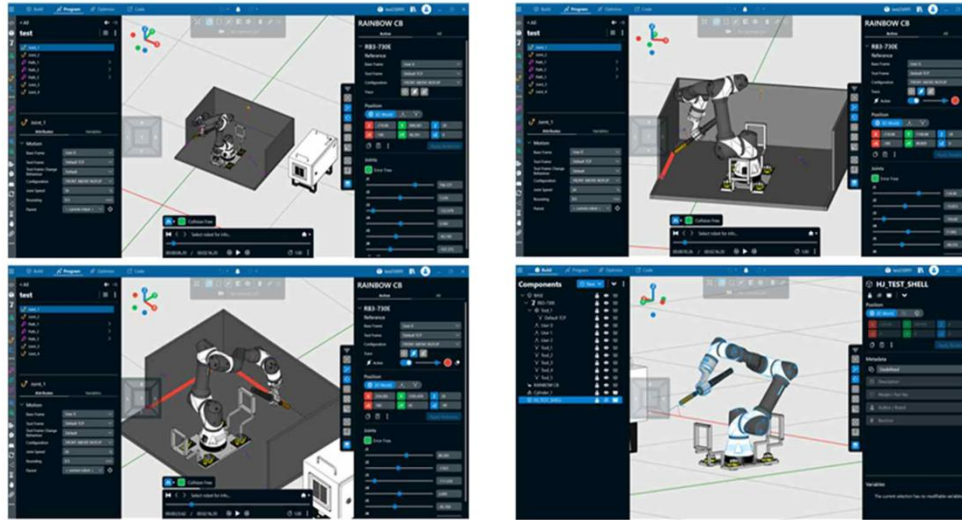
Automated Welding System for Ship Blocks (HJ S&C)



No	Category	Manufacturer	Model/Spec.	Utilization
1	Co-robot	RainbowRobotics	RB3-730ES	Welding Robot
2	Weld Monitoring	OTOS	WGC200	Vision Data Collection
3	Welding Machine	ESAB	ESAB 400i-Ra	
4	Control System	JCT		
5	OLRP/Monitoring	OCTOPUZ		
6	Welding Table	Customized		



Automated Welding System for Ship Blocks (HJ S&C)



- Modeling optimal robot DoF & welding conditions for required welding postures
 - Horizontal fillet welding
 - Vertical fillet welding
 - Horizontal curved fillet welding
- Simulation utilization for welding operation modeling to generate OLRP (Off-line Robot Programming)



- Fillet welding data generation and utilization
 - Analysis of weld leg length
 - Analysis of defects: Quality Assessment

Automated Welding System for Ship Blocks (HJ S&C)

- Lab-scale Execution
 - Generate OLRP after simulation using the OCTOPUZ tool



Welding Efficiency: 19.7 m/h of welding speed based on the same block



- Application Results(HJ S&C)
 - Achieved high-quality bead that eliminates the need for grinding



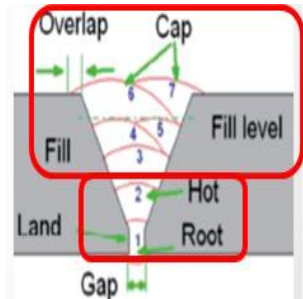
Automated Pipe Welding System (DAEYEON)

SS Welding Issues



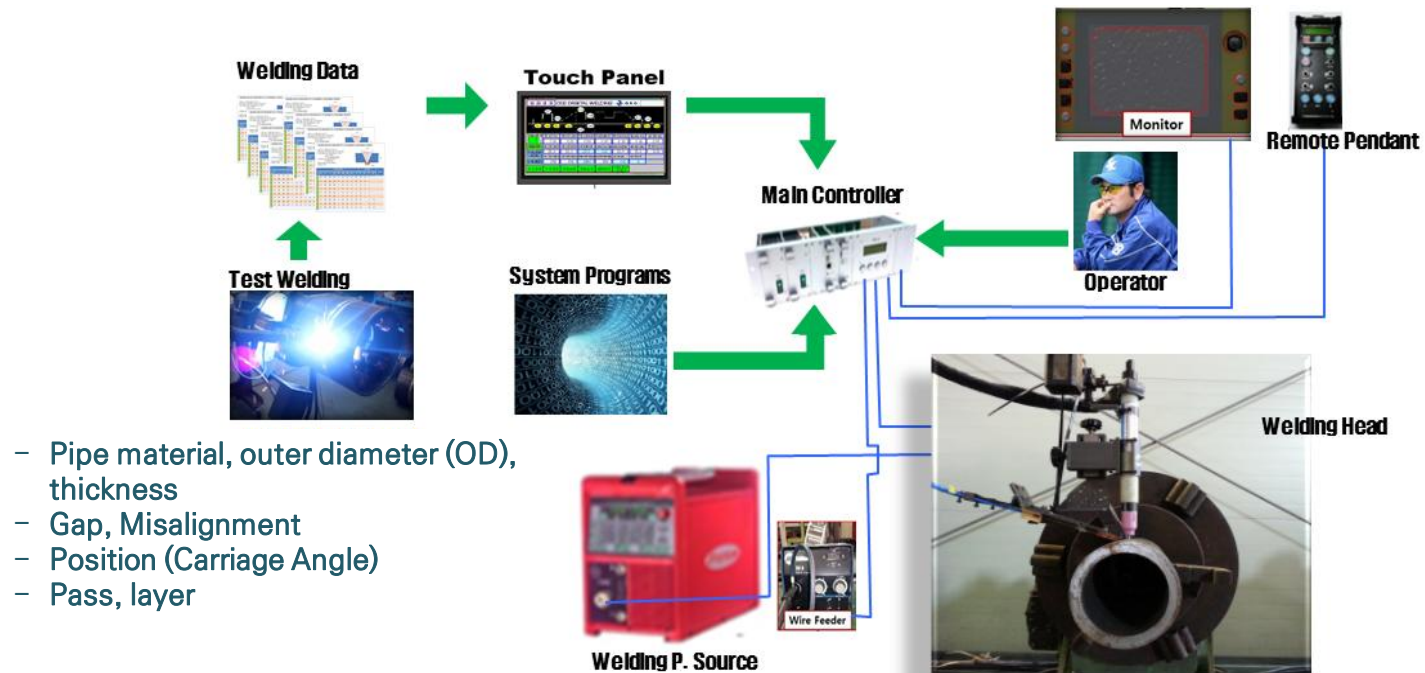
- Manual TIG welding being performed for all passes, regardless of the welding pass
- Low-efficiency welding method, despite eco-friendliness(minimal welding fumes & spatter)
- Productivity: 50-60 inches/day (Based on STD THK)
- Requires highly skilled welders

Issues

Root&Hot Pass Welding	Fill&Cap Pass Welding	Welding Spatter	Separation of Welding Process	Welding Groove Shape
			Transfer work for Fill & Cap Pass Welding after Root & Hot Pass Welding	
Manual TIG welding Lowest efficiency High-skill required	High efficiency, but poor environment (fumes & spatter)	Spatter removal via grinding causes noise and dust	Separated resources (personnel/space/equipment) & heavy material transfer	

Automated Pipe Welding System (DAEYEON)

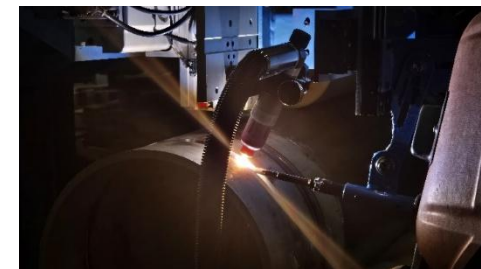
System Configuration



Fit-up Welding System



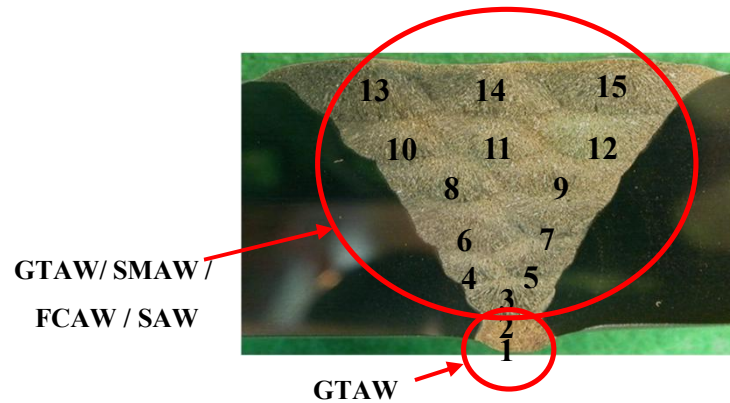
Control Unit & Welding Variables DB



Multi-Electrode Welding

Automated Pipe Welding System (DAEYEON)

Comparison of Before & After

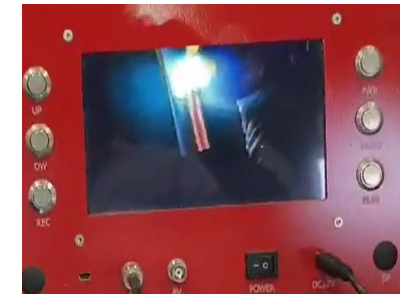
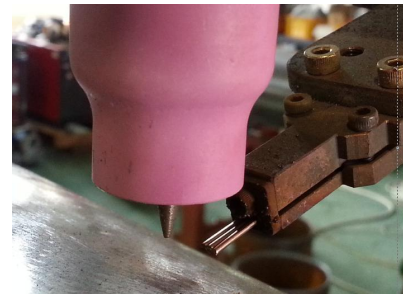


Category	Conventional Method (Combined Welding)					Developed Equipment (Single Process Welding)	
	Root Hot Pass	Fill/Cap Pass				Root Pass	Hot/Fill/Cap Pass
Welding Process	Manual GTAW	SMAW (Manual)	Auto GTAW	Semi-Auto FCAW	Auto SAW	Multi wire TIG	
						1 wire	2~3 wires
Deposition Rate (g/min)	10	22	18	65	100	10	50
Key Features	No internal slag	Low CAPEX	Low prod.	Spatter/fume/grinding	High prod./Not for slim pipes	High Arc-on Efficiency	High quality /Deposition rate

Automated Pipe Welding System (DAEYEON)

Results

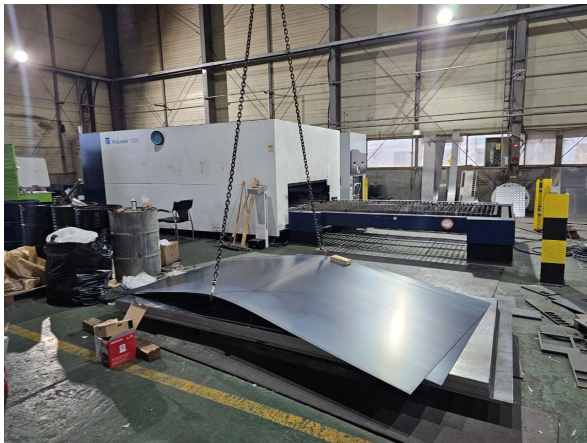
Category	Conventional Tech	Proposed Solution	Improvements
CS Pipe Welding	Manual GTAW + Semi-auto FCAW	Automated GTAW (Multi-electrode)	2x productivity
SS Pipe Welding	Manual GTAW	Automated GTAW (Multi-electrode)	3x productivity
Welding Fumes (CS)	195mg/m ³ (FCAW)	9mg/m ³ (GTAW)	Reduced to 1/20 ↓
Welding Spatter (CS)	Massive	Minimal	Improved working environment
Welding Personnel	Highly skilled welders	Equipment operators (1 operator per 2 units)	Resolves welder shortage issues
Welding Position	Close proximity	Remote control	Safe operations away from hazards
Material Transfer (CS)	Required	Not required	Optimized process & enhanced safety



Cutting Piece Sorting & Defect Prevention System (SUN-TECH)

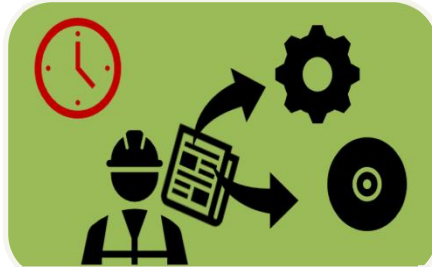
Background

- Various cut products(thickness & specifications) → Mixed-ordered(by companies) nesting issues
 - Manual sorting of cut pieces
 - Human errors caused by manual operations (e.g., misclassification, omissions)
- Thermal degradation of the cutting machine bed, leading to variations/inconsistencies in bed height

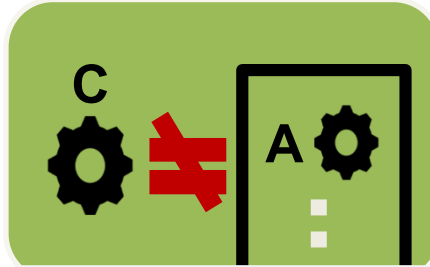


Cutting Piece Sorting & Defect Prevention System (SUN-TECH)

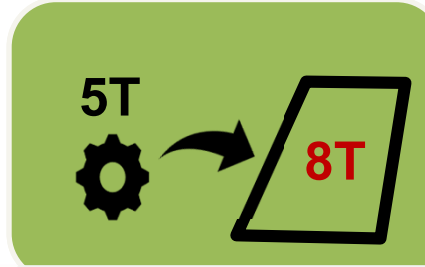
Project Objectives



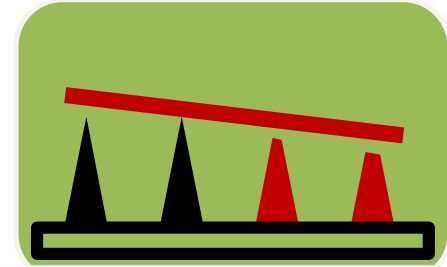
Manual drawing matching



Misclassification & Omission



Incorrect plate feeding



Cutting machine bed degradation

Real-time Classification & Defect Prevention System by CAD Matching

Computer Vision

AI drawing analysis & matching system

Real-time monitoring & alert system

IoT Interface

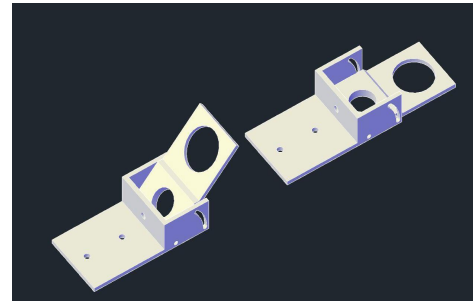
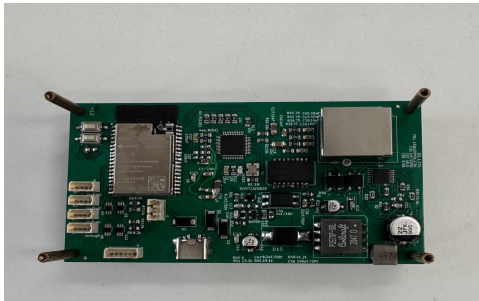
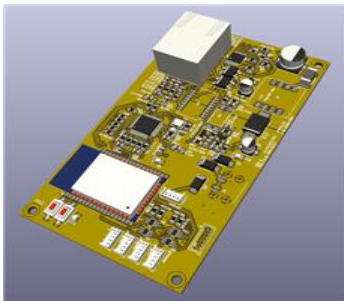
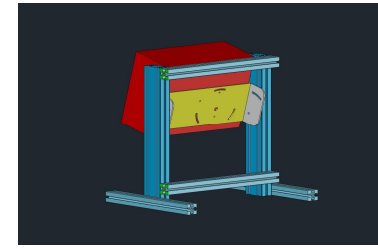
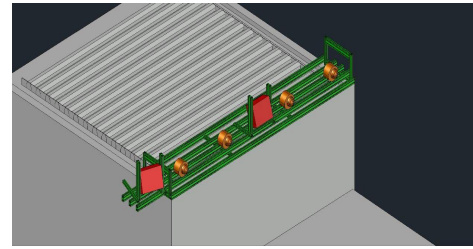
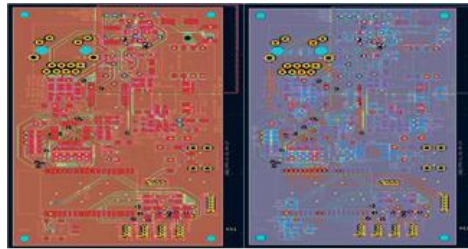
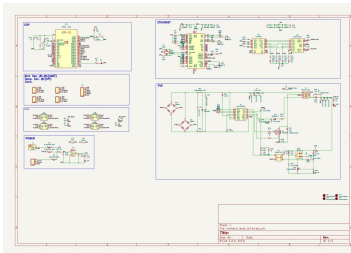
Metal plate thickness measurement

Cutting bed degradation detection

Cutting Piece Sorting & Defect Prevention System (SUN-TECH)

IoT Modules

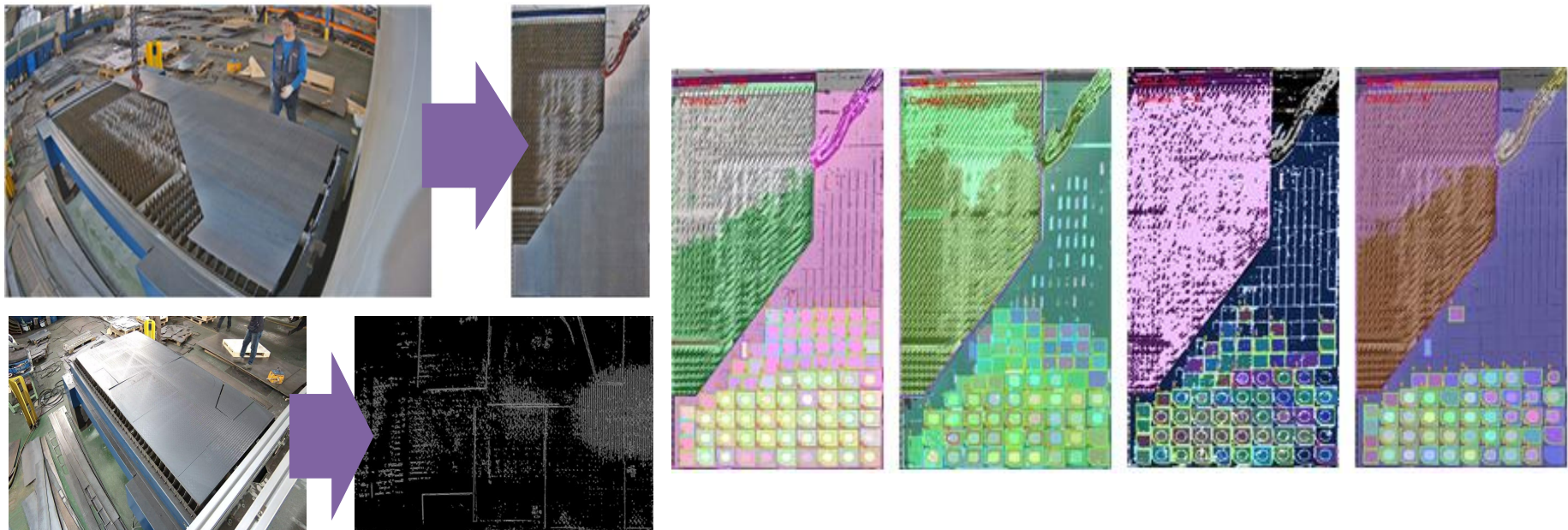
- Raw material thickness measurement – Prevents drawing-material mismatch
- Raw material tilt measurement – Detects degradation of laser cutting bed
- Raw material temperature measurement – Prevents operator burn injuries



Cutting Piece Sorting & Defect Prevention System (SUN-TECH)

AI-Based Drawing Matching

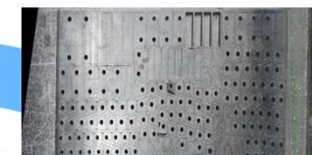
- Vision AI Model for Cut Materials
 - Background-segmented images → Determines shapes & Extracts outlines of materials
 - Extracts feature offsets (corners, edges, blobs, etc.) from images
 - Matching results between vision and drawing data → Alerts operators(Monitoring System)



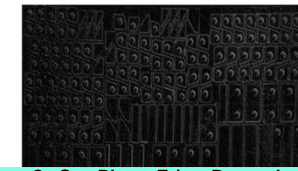
Cutting Piece Sorting & Defect Prevention System (SUN-TECH)

AI-Based Drawing Matching

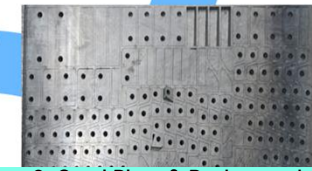
- Drawing Analysis & Matching Algorithm
 - Projection system to display drawing matching results
 - Alignment & Distortion calibration of target surfaces for projection mapping
 - Development of an intuitive UI to verify result data



1. illumination & Top-view Transformation



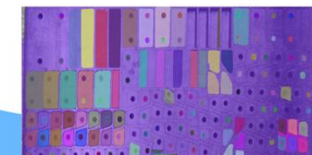
3. Cut Piece Edge Detection Algorithm



2. Steel Plate & Background Segmentation



4. Cut Piece vs Plate Matching Algorithm

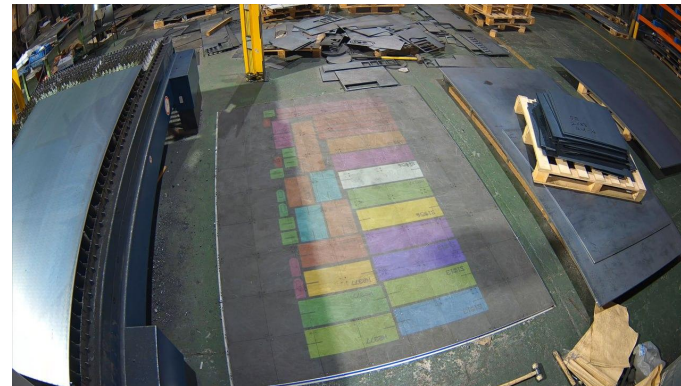


5. Visualization Algorithm

Cutting Piece Sorting & Defect Prevention System (SUN-TECH)

AI-Based Drawing Matching

- Test-bed of Cut Products Recognition System
 - Execute matching after vision recognition of cut products placed on the stage
 - Lighting systems, coatings, and patterning techniques to improve AI performance



Thank you for your attention.